

# MATTHEW POOL

## Software Engineer | Product Designer

UI/UX | Native Mobile | Full-Stack Web | AI/ML | B2B/B2C

 mathyou.me@gmail.com

 903-576-0223

 Winnsboro, TX

 matthew-pool.github.io

## PROFESSIONAL SUMMARY

I graduated Summa Cum Laude with a 3.99 GPA in Computer Science — backed by over a decade of combined customer service and technical support experience, plus five years as a top-ranked outbound B2B/B2C sales representative for Fortune 500 clients. This unique blend allows me to bridge the gap between user experience and technical implementation — managing the end-to-end product lifecycle from design to production deployment to build robust, scalable solutions that provide high ROI.

## EDUCATION

### Bachelor of Science in Computer Science

*Southern New Hampshire University | Sep 2021 – Dec 2024*

 **Summa Cum Laude • 3.99 GPA • President's List**

 **Certificate in Data Analysis • uCertify (2024)**

### Associate of Arts in General Studies

*Tyler Junior College | Aug 2003 – May 2005*

# TECHNICAL SKILLS

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## Product Design

UI/UX Design, Figma, Inkscape, GIMP, Adobe Creative Suite, DaVinci Resolve, CAD, SketchUp, Responsive Design, Motion Design, UI Animation, User Flows, Information Architecture

## Programming Languages

Python, JavaScript, TypeScript, Kotlin, Java, C/C++, C#, SQL

## Mobile Development

Android SDK, Jetpack Compose, Material Design 3, MVVM Architecture, Coroutines, Hilt, Firebase, REST APIs

## Frontend Web

React, Angular, RxJS, HTML5, CSS, SVG Animation, Vite, Bootstrap, Tailwind CSS

## Backend & Databases

Node.js, Express.js, Spring Boot, MySQL, MongoDB, Mongoose (ODM), AWS DynamoDB, Firebase Firestore

## Data Science & Machine Learning

Keras (TensorFlow), scikit-learn, Pandas, NumPy, Matplotlib, DNN, CNN, Reinforcement Learning (RL), Deep Q-Networks (DQN)

## Gaming & Graphics

Unity, OpenGL/GLSL, 2D/3D Graphics, Story-telling, Game Design & Development

## Testing & QA

Postman, pytest, JUnit, MockK, Mockito, Espresso, Robolectric, Jacoco, Jest, Selenium

## Infrastructure & Security

Docker/Docker Compose, AWS, CI/CD, Git/GitHub Actions, Markdown, Documentation, JWT, OWASP, Snyk, Gradle, Maven

# TECHNICAL PROJECTS

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|                                   |                                                                      |                                                                                                                                                                                                                                                         |
|-----------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Flick (Android)</b>            | <i>Kotlin, Jetpack Compose, MVVM, Hilt, Firebase, GitHub Actions</i> | Leveraged the Archive.org API to build a highly scalable, zero-infrastructure-cost video library. Paired this with a proprietary ML recommendation engine and asynchronous caching/preloading to deliver a low-latency, real-time streaming experience. |
| <b>Wormhole</b>                   | <i>Python, OOP, State Management</i>                                 | Architected a complex interactive Python terminal game utilizing advanced object-oriented programming, dynamic state management, and robust game loop logic.                                                                                            |
| <b>Full-Stack Travel Platform</b> | <i>Angular, TypeScript, Docker, AWS DynamoDB</i>                     | Engineered an Angular SPA with a Dockerized backend, orchestrating a full DB migration from MongoDB to AWS.                                                                                                                                             |
| <b>The Existence Paradox</b>      | <i>React, TypeScript, Tailwind CSS</i>                               | Interactive web application featuring dynamic state management and a complex decision-tree navigation system.                                                                                                                                           |
| <b>Financial Security Audit</b>   | <i>Java, Spring Boot, OWASP, AES-256</i>                             | Refactored a legacy REST API to implement encryption at rest, mitigating injection vulnerabilities to achieve OWASP compliance.                                                                                                                         |
| <b>Machine Learning Models</b>    | <i>Python, Keras (TensorFlow), scikit-learn, NumPy</i>               | Trained and tuned a CNN for image classification and a Deep Q-Network (DQN) for reinforcement learning pathfinding.                                                                                                                                     |
| <b>Product Design &amp; UX</b>    | <i>Figma, Wireframes, Prototyping</i>                                | End-to-end UX design for mobile and web apps: personas, wireframes, hi-fidelity interactive prototypes.                                                                                                                                                 |
| <b>Professional Portfolio</b>     | <i>HTML5, CSS3, Vanilla JS</i>                                       | Designed and hand-coded a zero-dependency responsive portfolio featuring a custom dark mode and complex SVG animations.                                                                                                                                 |

More projects and details at [matthew-pool.github.io](https://matthew-pool.github.io)

## PROFESSIONAL EXPERIENCE

### Lead Software Engineer & Product Designer

*AppBuddy (Independent Venture)*

*Jan 2025 – Present*

- **Mobile Product Strategy:** Architected and launched **Flick**, a native Android video streaming platform — designed in Figma, built in Kotlin (MVVM), and deployed to Google Play with a proprietary ML recommendation engine.
- **Backend & Logic Engineering:** Architected **Wormhole**, a Python-based application, utilizing advanced object-oriented programming (OOP), state management, and complex logic design.
- **Web & Frontend Architecture:** Engineered a zero-dependency personal portfolio platform utilizing pure HTML/CSS/JS, and built **Existence Paradox**, a React/TypeScript application featuring dynamic state management and complex decision-tree routing.
- **Development Velocity & Prototyping:** Leveraged generative AI toolchains to accelerate end-to-end development—rapidly iterating on high-fidelity UI assets, generating boilerplate code, and scaling solo output across all applications.
- **CI/CD & QA:** Built robust CI/CD pipelines via GitHub Actions automating unit (JUnit/MockK) and UI tests (Espresso) to ensure stable production deployments.

### Outbound Sales & Account Management (2004–2009)

★ Recurring Top Performer

**Sales Representative** • T.C.I.M. Services (B2B/B2C Outbound Sales)

- Conducted high-volume outbound sales calls and account management on behalf of Fortune 500 clients including American Express, AT&T, QVC, Citibank, Blue Cross Blue Shield, among others.
- Consistently ranked among the top performers in a rolling pool of 100+ representatives, exceeding KPIs across acquisition, upsell, and retention metrics.
- Developed call scripts adopted across the team and actively trained junior sales representatives on objection handling and closing techniques.

## Technical Support, IT Operations & Inbound Sales (2011–2013)

**Technical Support Specialist & Desktop Support** • Sutherland | New Electronics Warranty | TeleNetwork

- Diagnosed and resolved complex hardware, software, VoIP, and ISP issues via remote desktop tools; documented resolutions in CRM knowledge bases to improve team efficiency.
- Translated complex technical issues into clear guidance for diverse consumer audiences.
- Identified upsell and upgrade opportunities during support interactions, consistently meeting or exceeding inbound sales and satisfaction targets.
- Clients included Frontier Communications and DirecTV premium support customers.

## Shift Management & Customer Service (1997–2003 / 2009–2010)

 Management & Ops

**Shift Manager & Retail Associate** • Kmart / Fast Food & Customer Service (East TX)

- Directed workflow, delegated tasks, and trained new hires across fast-paced retail and food service environments.
- Regularly trusted to close stores independently at night, maintaining full financial accountability for cash registers and daily deposits.
- Managed inventory, restocking, and merchandising to optimize sales floor presentation.
- Operated cash registers and promoted store membership programs at store entrance.

## Independent Projects & Family Leave (2014–2021)

**Construction & Primary Caregiver**

- Developed woodworking skills while working at Diamond G Cedar (family-owned business) and doing independent research online.
- Applied project management skills to successfully design and construct a 3-bedroom residential home from an existing woodshed, minimizing project costs under strict budgetary and labor constraints by carefully salvaging and cleaning raw materials — one nail at a time.
- Served as caregiver for a family of four and helped facilitate spouse's successful return to college to receive her BBA in Finance, prior to returning to college myself to earn my Computer Science degree.